

Fall 2026: Rice D2K Capstone Project Proposal

Email your Project Proposal to d2k@rice.edu for review **before July**
Contact us ahead of time if you wish to engage in discussion regarding your proposal.

Project Title

Pollen identification using machine learning

Project Pitch

The Earth has been through periods of extremely warm climate in the geological past, when CO₂ levels were higher than present. We want to know how well computer simulations predict high-CO₂ climates of the future, and the only way we have to test those simulations is by seeing how well they predict the warm climates of the past. How do we test a simulation of climate from millions of years ago? Fossils! Specifically fossil pollen grains. (Yep, pollen fossilizes. Pollen is abundant and super resistant to degradation so there can be thousands of fossil pollen grains in a gram of rock or sediment.)

Plants are famously sensitive to climate. Arctic willows don't grow in Texas; palm trees don't grow in Alaska. We want to make a map of the flora of North America at one of the warmest times in geological history – the Eocene epoch about 50 million years ago. Once we have the map, we will compare it with advanced computer simulations of Eocene climate to see if the climate zones correspond with plant distributions the way they should.

The floral map of Eocene North America should provide a powerful test of our ability to simulate the hot Eocene climate, and thus improve our confidence in predicting hot climates of the future – something that is of profound scientific interest and societal importance as we learn to adapt to the effects of a higher CO₂ atmosphere.

Project Description

The biggest hurdle in making our floral map of Eocene North America is gathering a spatially dense set of samples. This is where the fossil pollen comes in. We have a collection of hundreds of microscope slides containing fossil pollen, and we are digitizing these slides with a scanning light microscope. An individual image has hundreds to thousands of pollen grains in it. We have a set of slide images that have already been tagged and identified by experts. Because we hope to use hundreds of samples to make our map, we need to greatly speed up the process of detecting and classifying pollen species in each image.

Project Objectives

Please provide bullet points outlining your project objectives.

- Develop a model to detect (if needed) and classify upto 30 different species of pollen

from multifocal digital microscopy images

- Develop an end-to-end pipeline for data loading, preprocessing, and AI/ML models
- Create a GUI that can be easily used by scientists and researchers with limited to no coding background for running inference using your models and finetuning on newer datasets.

Project Duration

One semester

Project Outcomes

- *End-to-end pipeline for data handling, preprocessing and AI/ML models*
- *Well-documented, reusable and reproducible codebase*
- *Technical report, presentation*

All code and models will be made openly available to support reproducible science. The Smithsonian Institution will retain ownership of the fossil image data and extracted palynomorph dataset. The analytical methods and code will be published under an open-source license to benefit the broader scientific community. We envision this work forming the foundation for future palynomorph analysis tools and potentially leading to publications in paleontology and computer vision venues.

Data Description

The training data consists of annotated pollen classes for up to 30 pollen species. Each slide image was acquired using multifocal microscopy, capturing the same field of view at different focal planes. This three-dimensionality is essential because palynomorphs are three-dimensional objects in which different sections of the grains are detected in focus at different depths along the z-axis.

The full dataset will comprises ~1,200 multifocal microscope slide images from the Smithsonian Institution's fossil collection. Each complete slide image exceeds 25GB in size due to the high resolution required for microscopy and the multiple focal planes captured per field of view. Collectively, the dataset represents ~1.75 petabytes of data. Test data will include full slide images (not just annotated subregions) with the goal of extracting 100% of palynomorphs present. Note: Complete annotations for full slides are not currently available but will be obtained during the first few weeks of the semester to enable proper validation.

Data Confidentiality

Answer the questions below about the data you will be providing for the project.

1. No, the data is not confidential. The Smithsonian Institution's fossil collection images are scientific data intended for research use. There is no protected health information (PHI) or privacy concerns associated with fossil specimens.
2. Data will be shared via Rice Box folder or a Smithsonian-controlled server with appropriate access credentials provided to the student team. Initial training data (85 annotated slides) will be transferred at project start, with arrangements made for processing the full collection as the project progresses.

3. This is a course at Rice University and students will need to present their work.

- a. Public presentations. Student teams will present their findings at the end of semester D2K Showcase in the form of a 1 minute elevator pitch video as well as a short presentation and poster session. No Confidential Information will be included in the presentations. The 1 minute elevator pitch video will also be posted publicly on the D2K website.

Do you consent to public presentation of this project?

☐ ☒ Yes

☐ ☐ No

- b. Internal presentations. In class, student teams will present and discuss their findings with instructors and with other students. No Confidential Information will be included in

the presentations.

Do you consent to the internal presentation and discuss of this project in a class setting?

☐ ☒ Yes

☐ ☐ No

C. Do you consent to public disclosure (title and one sentence description) of this project on the D2K website and social media?

☐ ☒ Yes

☐ ☐ No

Student Teams

Required Skills:

- *Strong programming skills in Python*
- *Experience with deep learning frameworks (PyTorch or TensorFlow)*
- *Computer vision and image processing knowledge*
- *Ability to work with large-scale datasets*

Preferred Skills:

- *Experience with object detection or segmentation models*
- *Familiarity with microscopy or biomedical imaging*
- *High-performance computing experience*
- *Background in biology, geology, or paleontology (helpful but not required)*

The project requires students who can handle computationally intensive tasks and are comfortable working with very large datasets that exceed typical computer memory limitations.

Project Type: Scientific Research / Climate Science / Paleontology	Student Restrictions: None
Technical Area(s): • Computer Vision • Deep Learning / Machine Learning • Image Processing • Large-scale Data Processing • Object Detection/Segmentation	Student Preferences: • Computer Science • Data Science • Electrical & Computer Engineering • Computational & Applied Mathematics • Statistics (with strong programming)
Skill Sets Required: • Deep Learning Preferred: • Computer Vision • Object Detection/Segmentation	

Sponsoring Organization

The Smithsonian Institution is a non-profit educational and research institution and museum complex. The Smithsonian maintains one of the world's largest collections of natural history specimens, including extensive fossil collections that document Earth's biological and climatic history. This project is affiliated with the Smithsonian's paleontology research program.

Sponsor Mentors

[To be provided by Smithsonian Institution - name, title, email, and phone number of primary mentor who will meet weekly with student team]

Scott Wing, Paleobotany Curator, wings@si.edu

Ingrid Romero, Postdoctoral Research Fellow, romeroic@si.edu

Sponsor Administrative, Communications & Financial Contacts

[To be provided by Smithsonian Institution - administrative, communications, and financial contact information]

Rice Faculty Mentors

Dr. Arko Barman

Data Science Education

Following the completion of the capstone project, would you be interested in D2K faculty instruction in data science and AI aligned with your organization and capstone project?

- ☒ ☐ Extremely interested - Highly enthusiastic about enhancing our team's skills in data science and AI and are eager to start as soon as possible.
AI and are eager to start as soon as possible.
- ☐ ☐ Very Interested - See great value in this opportunity and are strongly considering it.
- ☐ ☐ Slightly Interested – Have some interest but need more information to decide.
- ☐ ☐ Not interested - Do not see this as a fit for our team's needs at this time.

[To be completed by Smithsonian Institution regarding interest in follow-up AI education]

Is your project funded?

[To be provided by Smithsonian Institution if applicable]

The project has been funded by the Smithsonian Climate Change Fellowship: Our Shared Future. Life on a Sustainable Planet; the Smithsonian Office of the Undersecretary for Science; LSP Pathfinder Grant: Effects of Extreme global warming on the distribution on terrestrial plants. Currently is been funded by the NMNH Office of the Associate Director for Science and Chief Scientist; and the Bill and Dianne Ryan Family